YILDIZ TECHNICAL UNIVERSITY

Department of Metallurgical and Materials Engineering

Course: MSE4121 Production Laboratory
Experiment: Exp 3: Powder Metallurgy
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1. Objective

This laboratory study aims to observe and practice the three primary steps of powder metallurgy (P/M): powder characterization, cold pressing, and sintering. The relations between powder morphology, green density, and sintered density are analyzed [1].

2. Theoretical Background

Powder metallurgy is a manufacturing technique where metal powders are compacted into a desired geometry and subsequently heated (sintered) below their melting point to form a solid component [1]. Before compaction, powders are evaluated for their apparent (bulk) density and fluidity, which dictate mold filling behavior and production speed. During pressing, particles are mechanically interlocked, yielding a fragile "green compact." The final mechanical integrity is achieved through sintering, wherein solid-state diffusion occurs at elevated temperatures, transforming weak mechanical bonds into strong metallic bonds and reducing internal porosity [2].

3. Experimental Data & Results

The experiment utilized a standard flow funnel, a 25 cm^3 cylindrical cup, a 50 kp press, and a 900°C furnace. Raw data obtained during the laboratory session are summarized in Table 1.

Parameter / Process Step		Measurements (Trials 1, 2, 3)	Calculated Average
Powder Characterization	Apparent Density (mass / 25 cm^3)	79.66 g, 78.31 g, 78.82 g	**3.157 g/cm^3**
	Fluidity (Flow time for 50 g)	30 s, 32 s, 29 s	**30.33 s**
Pressing (Green State)	Geometric Density ($V \approx 1.857$ cm^3)	-	**6.620 g/cm^3**
	Archimedes Density	-	**6.704 g/cm^3**
Sintering (900°C, 5 min)	Final Density (Archimedes)	-	**6.715 g/cm^3**

4. Discussion

The powder characterization revealed an average apparent density of 3.157 g/cm^3, which perfectly falls within the standard theoretical range of 2.80–3.20 g/cm^3 for water-atomized steel powders [1]. However, the fluidity test yielded an average flow time of ~30 seconds. Since perfectly spherical powders typically flow in ~15 seconds, this prolonged duration confirms that the utilized powder possesses an irregular particle shape, causing higher inter-particle friction [2].

Regarding densification, a noticeable discrepancy exists between the geometrically calculated green density (6.62 g/cm^3) and the Archimedes measurement (6.704 g/cm^3). The fluid displacement method (Archimedes) is more reliable here, as geometric calculations fail to account for microscopic surface irregularities. Upon sintering at 900°C for just 5 minutes, the density slightly increased to 6.715 g/cm^3. This minor densification proves that, despite the short duration and lower-bound temperature, inter-particle necking initiated, successfully shrinking pores.